TUTOR GUIDE • LEVEL 1 • TERM 1 • LESSON 6 OF 10

Tens Jump, Ones Hop



Big idea: Add and take away tens without carrying. A jump of ten changes only the tens digit. A hop of one changes only the ones digit. For a bigger step, jump the tens first, then hop the ones.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Addition and Subtraction (Mental calculation: a 2-digit number plus or minus tens, without renaming (syllabus 2.7); A 2-digit number plus or minus ones, without renaming; Number line jumps as a picture of adding and taking away) Strand label for the tutor only. Term 1 goes beyond the school Primary 1 syllabus on purpose; it is not MOE-aligned and must not be called that on child or parent pages. In this lesson Start here stays inside P1 mental calculation of a 2-digit number plus or minus tens or ones without renaming (syllabus 2.7), explored first with bundles, bars and number lines. Think harder (34 + 25, 68 - 24) and Go further (two jump sizes, a chain of jumps) go beyond P1 by splitting a 2-digit step into a jump and a hop. The owner or a tutor should check the placement against the syllabus before the first live class.
Objectives	• Add and take away a multiple of ten from a 2-digit number in your head, knowing that only the tens digit changes (34 + 20 = 54, 54 - 20 = 34). • Add and take away a few ones without passing a ten (34 + 3 = 37, 58 - 3 = 55), knowing that only the ones digit changes. • Split a step such as 34 + 25 into a jump of tens and a hop of ones, and say where you land after each part. • Draw a number line with jumps and hops, and use it to explain an answer.
Joining cold	**Remember warm-up covers:** Count in tens from any number (3, 13, 23 ...); Read a 2-digit number as tens and ones (46 is 4 tens and 6 ones); Count on a few ones from a tens number (40 and 5 more is 45). Only counting to 20 and showing a number with fingers or a picture. Nothing from an earlier Term 1 lesson is needed except what the warm-up re-teaches with bars and squares. A child who did the holiday course will know counting in tens. Lessons 5 to 7 are among the hardest to miss in Term 1 (plan, section 2). A child who misses Lesson 6 watches the recording and does workbook pages 1 and 2 before Lesson 7, which uses jumps to find a gap.
Child materials	• The tens bars and ones squares from the kit page (workbook page 6), cut out by a grown-up. They are the same kit as Lesson 1, so keep them in an envelope. • About 30 small things such as pasta, buttons or paper clips (only for the home practice; optional in the lesson) • The printed workbook (pages 1 to 5) and the kit page • A pencil, a rubber and spare paper
Tutor materials	• A slide deck and the kit bars and squares to show on camera if you wish • A visible clock or a timer slide No ruler is needed in the lesson; a child who has one at home may use it to draw straight number lines. Nothing needs to be bought. No matches, toothpicks or anything sharp. Children need a little clear floor space for the movement break; small things never go near the mouth or ears.
Tiers	Start here (7 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.

Contest-style move	Used at Think harder only: split the second number into tens and ones (jump, then hop); the order of the steps does not change where you land; fewest jumps with two jump sizes. Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first. Keep a paper number line ready to draw on camera as you talk.
- Tell parents a week before: print workbook pages 1 to 6, find the kit bars and squares from Lesson 1 (or cut out page 6 again), and a pencil and spare paper.
- Have the slides open. The slides show number lines with jump arcs; draw the arcs by hand on paper as well.
- Check the recording settings exclude children's video.

Safeguarding and recording (this lesson)

- Small things stay on the table: nothing goes in the mouth or ears. No matches, toothpicks or anything sharp. The kit is paper, so there is nothing to swallow.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Sneak-peek and exit answers are shown on the child's own paper to the camera at the same moment, once. There is no private chat and no board held up in front of the group.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. A child may march on the spot or stay seated.
- Never read out a child's name with a right or wrong answer. Never show a child's page in the recording; record the tutor and the screen only, with children's video excluded.
- A child who ticks I am still stuck gets a check-in in the open session next lesson (a quick visit to the shared room or a hand signal, parent nearby). No private messages to children.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: Tens Jump, Ones Hop	1:34
2–8	**Remember warm-up**	6	3: Warm-up: count in tens, Tens and ones, Ten and some more	5:24
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:25
12–28	**Explore**	16	5: Explore: jump with bars, Explore: hop with squares, Explore: jump and hop, Draw the jumps and hops, What did we notice?	12:50
28–38	**Mini-lesson**	10	8: Jump forward: add tens, Jump back: take away tens, Hop: ones, Jump, then hop: add, Jump, then hop: take away, Your turn: write the answer, Check: write the answer, Jump or hop?	8:27
38–42	**Brain break**	4	1: Jump the tens, hop the ones	3:38
42–70	**Workbook**	28	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	26:28
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	5:15
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:32
82–85	**Look-back**	3	1: Look-back	2:28
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:38

	Total	**90**		**74:38**

The script needs 74:38 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 88:33. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 42 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 42, skip in this order: optional items and extra quick questions first, then the second explore round, then any quick-fire in the mini-lesson, then the optional slide. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:34 of 2:00

0:00	SLIDE	Show slide **Tens Jump, Ones Hop** (s-title)
0:25	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your kit of bars and squares. Pause or rewind any time.*
0:38	SAY	Good morning, inquirers! Cameras on, pencil ready, and your kit of bars and squares on the table. This lesson is recorded so a friend who is away can catch up. We record me and the screen only, and you may switch your camera off whenever you like.
1:01	SAY	Picture a frog on a path of numbers. It can take a big jump of ten, or a small hop of one. Today we plan the frog's journey, and we find out which digit it changes.
1:19	DO	Greet each child by first name and check their camera shows the page and hands. [0:15]
	NOTE	You know the routine from Lesson 1, so keep this to two minutes. Tell any new child that recordings show only you and the screen. The frog is only a story to hook the idea: the jumps and hops are done with the kit.

2:00 – 8:00 Remember warm-up

script 5:24 of 6:00

2:00	SLIDE	Show slide **Warm-up: count in tens** (s-rem1)
2:25	REC	*Recording: join in! Say the numbers out loud. Pause if you need longer.*
2:32	SAY	Stand up if you can. We clap on every number. Count in tens with me: ten, twenty, thirty, all the way to one hundred.
2:44	CHILDREN	Count in tens together, clapping, to 100. [0:35]
3:18	SAY	Now a new start. We begin at three and add ten each time: three, thirteen, twenty-three. Point to the tiles as we say them.
3:30	CHILDREN	Count on from 3 in tens together: 3, 13, 23 … 93. Point at each tile as it is said. [0:40]
4:10	ASK	Look at the numbers. What stays the same every time? Tell your screen. [wait about 20 s for answers]
4:37	SLIDE	Show slide **Tens and ones** (s-rem2)
5:02	SAY	Tens and ones. A bar is a ten. A square is a one. Four bars and six squares is forty-six.
5:12	ASK	Fifty-two. How many tens? How many ones? [wait about 15 s for answers]
5:30	ASK	Seventy. How many tens? How many ones? [wait about 15 s for answers]
5:49	ASK	Eighty-three. How many tens? How many ones? [wait about 15 s for answers]
6:08	SLIDE	Show slide **Ten and some more** (s-rem3)
6:32	SAY	Ten and some more. Forty and five more is forty-five. We say the tens, then count on the ones.
6:42	ASK	Thirty and four more? [wait about 12 s for answers]
6:56	ASK	Fifty and seven more? [wait about 12 s for answers]
7:10	ASK	Eighty and two more? [wait about 12 s for answers]
	IF STUCK	If a child hesitates, say the tens word slowly (thirty… four) and point to the bars and then the squares in the picture. Do not move on until most children have answered: counting on from a tens number is what the lesson uses.

	NOTE	Answers: the ones digit stays 3; then 5 tens 2 ones, 7 tens 0 ones, 8 tens 3 ones; then 34, 57, 82. Children answer with fingers, hands or a whispered number. Nobody writes. Joining cold: all you need is counting to 20 and the picture.

8:00 – 12:00 Sneak-peek puzzles

script 3:25 of 4:00

8:00	SLIDE	Show slide **Sneak-peek puzzles** (s-sneak)
8:25	REC	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:34	SAY	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:47	READ	Puzzle one. Mei has the sticks in the picture. Then she gets another bundle of 10 sticks. How many sticks does Mei have now? You may draw.
9:02	CHILDREN	Children count and write the number on their own paper. [0:50]
9:52	READ	Puzzle two. Ravi says forty-three plus twenty is forty-five. Is Ravi right? Circle Yes or No. Then write how you know.
10:03	CHILDREN	Children write or draw their answer. [0:50]
10:53	HOLD UP	Everyone holds their paper to the camera at the same moment, once, after puzzle two. Do not read answers aloud. No private chat.
11:18	SAY	Thank you. We will come back to puzzle two in a few minutes.
	NOTE	Puzzle 1 answer: 36 sticks (26 and one more bundle). Puzzle 2 answer: no, 43 + 20 = 63; Ravi added 2 to the ones digit. Look for good ideas to praise later (a child who drew bundles, or who said 20 is 2 tens). Do not write scores or keep a list.

12:00 – 28:00 Explore

script 12:50 of 16:00

12:00	SLIDE	Show slide **Explore: jump with bars** (s-explore1)
12:25	REC	*Recording: pause each time I ask. Build the number with your bars and squares, then press play to check.*
12:34	SAY	Today the kit is a number path in your hands. A bar is a big jump of ten. A square is a small hop of one. Build thirty-four: three bars and four squares.
12:51	CHILDREN	Children build 34 with bars and squares. Tutor scans the cameras. [0:35]
13:26	HOLD UP	Everyone holds up their build to the camera on a count of three.
13:41	SAY	Take a big jump forward: add one bar.
13:45	ASK	What number do you have now? [wait about 12 s for answers]
14:00	SAY	Jump again.
14:01	ASK	And now? [wait about 10 s for answers]
14:12	SAY	Now jump back: take one bar away.
14:16	ASK	What number now? [wait about 10 s for answers]
14:27	SAY	Jump back two more times.
14:30	ASK	And now? [wait about 10 s for answers]
14:40	ASK	Which digit changed each time? Which digit stayed the same? [wait about 30 s for answers]
15:16	SLIDE	Show slide **Explore: hop with squares** (s-explore2)
15:40	SAY	Now the small hops. Start with twenty-four: two bars and four squares. A hop is one square. Hop forward three times: add three squares.
15:52	ASK	What number now? [wait about 12 s for answers]
16:06	SAY	Hop back two: take two squares away.

16:10	**ASK**	And now? [wait about 10 s for answers]
16:20	**ASK**	Which digit changed this time, the tens or the ones? [wait about 20 s for answers]
16:46	**SAY**	A jump changes the tens digit. A hop changes the ones digit.
16:52	**SLIDE**	Show slide **Explore: jump and hop** (s-explore3)
17:16	**SAY**	Now jump and hop together. Build thirty-four again. We add twenty-five. Jump twice, then hop five times. Hold up your kit when I count to three.
17:30	**CHILDREN**	Children add 2 bars then 5 squares. Hold up on a count of three. Answer 59. [0:50]
18:20	**SAY**	Next: forty-one plus thirty-eight. Jump three times, hop eight times.
18:24	**CHILDREN**	Build 41, add 3 bars and 8 squares. Answer 79. [0:42]
19:06	**SAY**	Now take away. Build sixty-eight. Take away twenty-four: jump back twice, hop back four times.
19:14	**CHILDREN**	Build 68, take away 2 bars and 4 squares. Answer 44. [0:42]
19:56	**SLIDE**	Show slide **Draw the jumps and hops** (s-explore4)
20:21	**SAY**	Now we put the same journey on paper. This is a number line. Draw a dot and write thirty-four. Draw a big arc for the jump of twenty, and write plus twenty. Write where you land.
20:39	**CHILDREN**	Children draw the dot, the arc and the landing number 54 on paper. Tutor draws on camera at the same time and shows the paper. [1:00]
21:39	**SAY**	Now a small arc for the hop of five. Write plus five, and write where you land.
21:48	**CHILDREN**	Children finish the line: 34, +20, 54, +5, 59. Check cameras. Then redo it for 68 take away 24 with back arcs: 68, -20, 48, -4, 44. [1:15]
23:02	**ASK**	Look at your line. Which arc is big and which is small? Why? [wait about 20 s for answers]
23:29	**SLIDE**	Show slide **What did we notice?** (s-explore5)
23:54	**ASK**	What did you notice? Which one changes the tens, a jump or a hop? Does it matter if you hop first or jump first? [wait about 30 s for answers]
24:36	**SAY**	A jump of ten changes the tens digit. A hop of one changes the ones digit. Today we always stay inside a ten: we never go past nine ones.
	IF STUCK	A child who adds the 2 of 25 to the ones: ask how many squares one bar is worth, then rebuild with 2 bars (the two jumps) and 5 squares (the five hops).
	IF STUCK	A child with too few bars: they may use 9 bars and 20 squares from the kit page; no more than that is needed in this lesson.
	NOTE	Main room only: no breakout rooms, so you can see every camera. Keep the picture of jump and hop as ten and one. Never say that you carry or rename; if a child asks what happens when the ones pass nine, say: good question, we use make-a-ten for that (Lesson 5).

28:00 – 38:00 Mini-lesson

script 8:27 of 10:00

28:00	**SLIDE**	Show slide **Jump forward: add tens** (s-m1)
28:25	**REC**	*Recording: keep watching and copy the jumps and hops onto your paper. Rewind if you miss a step.*
28:34	**SAY**	Here is the number path. We start on thirty-four and take a big jump of ten: we land on forty-four. Another jump lands on fifty-four. Two jumps of ten is twenty, so thirty-four plus twenty is fifty-four.
28:52	**ASK**	Which digit changed? Which digit stayed the same? [wait about 20 s for answers]
29:16	**SLIDE**	Show slide **Jump back: take away tens** (s-m2)
29:42	**SAY**	Jumping back takes tens away. From fifty-four, jump back ten: forty-four. Jump back again: thirty-four. Fifty-four take away twenty is thirty-four.
29:52	**ASK**	Now you try. Seventy-eight take away thirty. Which way is the jump? [wait about 20 s for answers]
30:18	**SLIDE**	Show slide **Hop: ones** (s-m3)

30:43	**SAY**	Now the small hops. A hop is one. Thirty-four plus three: hop three times to thirty-seven. The tens digit does not change. Thirty-seven take away two: hop back two to thirty-five.
30:58	**ASK**	Is five a jump or a hop? [wait about 12 s for answers]
31:14	**SLIDE**	Show slide **Jump, then hop: add** (s-m4)
31:39	**SAY**	Jump, then hop. Thirty-four plus twenty-five. Twenty-five is two tens and five ones. First jump two tens: thirty-four to fifty-four. Then hop five: fifty-nine.
31:51	**ASK**	How many tens are in thirty-six? How many ones? [wait about 15 s for answers]
32:10	**SLIDE**	Show slide **Jump, then hop: take away** (s-m4b)
32:36	**SAY**	Take away works the same way. Fifty-nine take away twenty-five. Jump back two tens: thirty-nine. Hop back five: thirty-four.
32:45	**ASK**	What is eighty-seven take away thirty-two? Say the jump first, then the hop. [wait about 25 s for answers]
33:16	**SLIDE**	Show slide **Your turn: write the answer** (s-m5)
33:42	**READ**	Your turn. Write the answers on your paper. Forty-five plus thirty. Seventy-eight take away twenty. Sixty-four take away three. Fifty-two plus thirty-six.
33:54	**CHILDREN**	Children write four answers. [1:05]
34:58	**SLIDE**	Show slide **Check: write the answer** (s-m5a)
35:24	**READ**	Seventy-five. Fifty-eight. Sixty-one. Eighty-eight.
35:26	**SLIDE**	Show slide **Jump or hop?** (s-m6)
35:51	**SAY**	Quick-fire. Is it a jump or a hop? Thirty.
35:55	**ASK**	Six? [wait about 10 s for answers]
36:06	**ASK**	Fifty? [wait about 10 s for answers]
36:16	**ASK**	Two? [wait about 10 s for answers]
	NOTE	Key words to use every time: jump (ten), hop (one), tens digit, ones digit. Do not teach carrying or renaming. Every example stays inside a ten: 34 + 25 uses 4 + 5 = 9 ones. Draw the number line on a paper or whiteboard as you talk so a child watching the recording sees the arcs.
	IF STUCK	If a child writes 35 for 45 + 30 or similar: ask how many bars 30 is, and whether a hop or a jump is needed. Build it with the kit and say the tens first.

38:00 – 42:00 Brain break

script 3:38 of 4:00

38:00	**SLIDE**	Show slide **Jump the tens, hop the ones** (s-break)
38:25	**REC**	*Recording: stand up and join in! Big jumps for tens, small hops for ones.*
38:32	**SAY**	Stand up and get a sip of water. Find a little space. We start on thirty-four. A big jump for every ten, a small hop for every one. Say the number each time.
38:48	**CHILDREN**	Call the moves one at a time and wait for everyone: jump twice (44, 54), hop three times (55, 56, 57), jump back once (47), hop back four times (46, 45, 44, 43). Count the numbers aloud together. [2:50]

42:00 – 70:00 Workbook

script 26:28 of 28:00

42:00	**SLIDE**	Show slide **Workbook page 1: Start here** (s-ws1)
42:25	**REC**	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
42:34	**READ**	Page 1, Start here. Question 1: Mei gets one more bundle of ten. Fill in the bundles, the singles and the number. Question 2: draw 3 tall bars of ten and 4 single sticks, then 2 more tall bars. Question 3: jump forward. Twenty-three plus ten. Twenty-three plus forty. Fifty-one plus thirty. Seven plus sixty.
43:04	**CHILDREN**	Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [9:00]

52:04	**SLIDE**	Show slide **Check page 1** (s-ws1ans)
52:29	**READ**	Thirty-six. Fifty-four. Thirty-three, sixty-three, eighty-one, sixty-seven. Mark your own page in another colour.
52:36	**SAY**	After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
52:45	**SLIDE**	Show slide **Workbook page 2: Start here** (s-ws2)
53:10	**READ**	Page 2. Question 4: jump or hop. Thirty-four take away ten. Fifty-eight take away twenty. Fifty-eight take away three. Sixty-two plus seven. Question 5: Ali has 68 pencils and gives away 3 boxes of ten. Question 6: Ben says forty-five plus twenty is forty-seven. Is Ben correct? Circle Yes or No. Then fill in the blanks. Question 7: start at 36 and draw two jumps of ten.
53:47	**CHILDREN**	Page 2, questions 4 to 7. [9:00]
62:47	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
63:12	**READ**	Question 4: twenty-four, thirty-eight, fifty-five, sixty-nine. Question 5: thirty-eight. Question 6: no, sixty-five. Question 7: fifty-six.
63:20	**SLIDE**	Show slide **Choose your next page** (s-choice)
63:45	**SAY**	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time.
63:58	**CHILDREN**	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. [4:30]
	IF STUCK	Q2 (drawing): draw each ten as one tall bar, like the bars in the kit. Q3 and Q4: ask is it a jump or a hop, then say the tens first. Q6: build 45 with the kit and add 2 bars. Q7: draw a dot for 36, then an arc and write +10.
	IF STUCK	A child who is still stuck ticks I am still stuck on the look-back page. You check in during the open session in the next lesson (a quick visit to the shared room or a hand signal, parent nearby). Never a private message.
	NOTE	Think harder is for ideas taught today in a new setting (jump and hop together). Go further is for mastery and extension. Do not announce who is on which page. Pairs are never split unless both children agree.

70:00 – 76:00 Talk time

script 5:15 of 6:00

70:00	**SLIDE**	Show slide **Talk and explain** (s-talk)
70:25	**REC**	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*
70:32	**READ**	Mei says: when you jump forward 2 tens, the ones digit changes. Is Mei right? Use your kit to show why.
70:43	**CHILDREN**	Build 34 and jump forward 2 tens with the kit. Which digit changed? [0:50]
71:33	**CHILDREN**	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [1:00]
72:33	**ASK**	Who would like to share? Volunteers first. A pass is always fine. [wait about 90 s for answers]
74:09	**SLIDE**	Show slide **Back to the sneak-peek** (s-sneakans)
74:34	**SAY**	Remember puzzle two? Ravi said forty-three plus twenty is forty-five. Twenty is two tens, so we jump, we do not hop. Forty-three plus twenty is sixty-three.
74:47	**ASK**	Did anyone change their mind about puzzle two while we worked? You can just nod. [wait about 20 s for answers]
	NOTE	Model answer: no. 34 + 20 = 54: the 4 ones are still there. Two tens are two bars, so only the tens digit changes (3 to 5). Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:32 of 6:00

76:00	SLIDE	Show slide **Exit ticket** (s-exit)
76:25	REC	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:33	SAY	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.
76:41	CHILDREN	Children answer E1 to E3 on their own. Remind them they may draw the jumps. [2:50]
79:31	HOLD UP	Everyone holds their ticket to the camera at the same moment so only you see it. No private chat.
79:56	SLIDE	Show slide **Check the exit ticket** (s-exitans)
80:21	READ	Seventy-six. Forty-four and seventy-one. Fifty-nine.
80:24	SAY	Check your own ticket in a different colour. Well done. I will stamp your inquiry journal.
	NOTE	Mark S, D or H in the tutor tracker (tutor only): S = secure on all three, D = developing (one wrong), H = needs help (two or three wrong). If D or H, plan two minutes of catch-up at the start of Lesson 7, in the open session, no names. Do not read out any child's result.

82:00 – 85:00 Look-back

script 2:28 of 3:00

82:00	SLIDE	Show slide **Look-back** (s-look)
82:25	REC	*Recording: pause and fill in your look-back box. You can pass.*
82:30	SAY	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out.
82:48	CHILDREN	Children fill in the look-back box on page 5. [1:40]

85:00 – 90:00 Flex time and goodbye

script 0:38 of 5:00

85:00	SLIDE	Show slide **Well done, inquirers!** (s-end)
85:25	SAY	Well done, inquirers. Next lesson is Take Away, Count Up. You need your kit, a pencil and paper. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: what if?** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: start at 5 and jump in tens. What is the last number you can reach before one hundred? How many jumps is that?
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: 95, after 9 jumps (5, 15, 25 … 95); one more jump would pass one hundred. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask How do you know? and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Is it always true?	4 (optional)	Mei says: when you add a ten, only the tens digit changes. Try 34, 67 and 90 with your kit. What do you notice?	*Recording: pause. Try adding ten to 34, 67 and 90. What do you notice?*	For 34 and 67 only the tens digit changes: 44 and 77. For 90 + 10 the answer is 100, a new number with a hundreds digit (a later lesson).
Make a puzzle for your grown-up	4 (optional)	Pick a start number. Write one jump and one hop, like start at 23, jump 3 tens, hop 4. Give it to your grown-up to solve.	*Recording: pause. Write a jump and hop puzzle for a grown-up to solve.*	No single answer. Check that the puzzle has a start number, a jump of tens and a hop of ones that stays inside a ten. Example: 23 + 30 + 4 = 57.
Teach it back	3 (optional)	Pretend your grown-up or a toy has never seen jumps and hops. Teach them how to work out 52 + 31.	*Recording: pause. Teach a grown-up how to work out 52 + 31 with a jump and a hop.*	31 is 3 tens and 1 one. Jump 3 tens: 52 to 82. Hop 1: 83.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): Mei has the sticks in the picture. Then she gets another bundle of 10 sticks. How many sticks does Mei have now? You may draw.

> **Answer: 36**

Puzzle 2 (think): Ravi says 43 + 20 = 45. Is Ravi right? Circle Yes or No. Then write how you know.

> **Answer: No. 43 + 20 = 63. 20 is 2 tens, so the tens digit changes, not the ones.**

Unmarked. Papers are shown to the camera once, together. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **These are Mei's sticks. She gets another bundle of 10 sticks. Fill in the blanks.**



(a) Now Mei has ___ bundles of ten.
(b) She also has ___ single sticks.
(c) Mei has ___ sticks altogether.

> **Answer: 3 bundles, 7 singles, 37 sticks**

Hint (say this first): Count the bundles in tens: 10, 20, 30. Then count on the 7 singles.
Worked solution: The picture shows 2 bundles and 7 singles, which is 27. One more bundle makes 3 bundles, and the 7 singles stay. 30 and 7 is 37.
Common mistakes: Adding the bundle to the singles (28).; Forgetting the old bundles (17).
Time for a middle child: about 2 min.

S2 **In the box, draw 3 tens as 3 tall bars and 4 single sticks. Next, draw 2 more tall bars. How many sticks are there altogether?**

There are ___ sticks altogether.

> **Answer: 54 (5 bars drawn and 4 singles)**

Hint (say this first): A bundle of ten is a tall bar, like the bars in your kit. Draw 3 bars, then 4 short lines, then 2 more bars.
Worked solution: 3 bars and 4 singles is 34. Two more bars add 20, so there are 5 bars and 4 singles: 54.
Common mistakes: Adding the 2 to the singles (36).; Drawing 2 sticks instead of 2 bars of ten.
Time for a middle child: about 3 min.

S3 **Jump forward. Fill in the blanks.**

(a) 23 + 10 = ___
(b) 23 + 40 = ___
(c) 51 + 30 = ___
(d) 7 + 60 = ___

> **Answer: 33, 63, 81, 67**

Hint (say this first): Think of the tens as bars. 23 + 10 is one more bar. The ones digit stays the same.
Worked solution: (a) 2 tens and 1 ten make 3 tens: 33. (b) 2 tens and 4 tens make 6 tens: 63. (c) 5 tens and 3 tens make 8 tens: 81. (d) 7 is 7 ones; add 6 tens to get 67.
Common mistakes: Changing the ones digit (23 + 40 = 27).; (d) writing 13 (7 + 6).
Time for a middle child: about 3 min.

S4 **Jump or hop. Fill in the blanks.**

(a) 34 - 10 = ___
(b) 58 - 20 = ___
(c) 58 - 3 = ___
(d) 62 + 7 = ___

Answer: 24, 38, 55, 69

Hint (say this first): A tens number is a jump. A ones number is a hop. Say which one it is before you work it out.
Worked solution: (a) 3 tens take away 1 ten is 2 tens: 24. (b) 5 tens take away 2 tens is 3 tens: 38. (c) 3 is a hop: 8 ones take away 3 is 5 ones: 55. (d) 7 is a hop forward: 2 and 7 make 9 ones: 69.
Common mistakes: Taking the 3 from the tens digit (58 - 3 = 28).; Hopping the wrong way in (d).
Time for a middle child: about 3 min.

S5 **Ali has 68 pencils. He gives 3 boxes of 10 pencils to his friend. How many pencils does Ali have left?**

There are ___ pencils left.

Answer: 38 pencils

Hint (say this first): 3 boxes of ten is 30 pencils. Jump back 3 tens from 68.
Worked solution: 3 boxes of ten is 30 pencils. 68 - 30 = 38: jump back 3 tens (58, 48, 38). The 8 ones stay.
Common mistakes: Taking away 3 (65).; Adding instead of taking away (98).
Time for a middle child: about 2.5 min.

S6 **Ben says 45 + 20 = 47. Fill in a to c.**

(a) Is Ben correct? Circle Yes or No.
(b) The right answer is ___.
(c) Ben added 2 ___ by mistake. He should add 2 ___.

Answer: No. The right answer is 65. He added 2 ones instead of 2 tens.

Hint (say this first): Build 45 with the kit. Is 20 two bars or two squares?
Worked solution: 20 is 2 tens, so Ben should jump two tens: 45, 55, 65. He added 2 ones and got 47. A jump changes the tens digit.
Common mistakes: Saying he is correct because 20 has a 2 in it.; Not saying what he mixed up (any clear sentence about tens and ones is fine).
Time for a middle child: about 2.5 min.

S7 **Start at 36. On a number line in the box, draw 2 jumps of ten. Where do you land?**

I land on ___.

Answer: 56 (two jumps of ten from 36: 46, then 56)

Hint (say this first): Draw a dot for 36. Draw an arc to the right and write +10. Do it once more.
Worked solution: 36 + 10 = 46, then 46 + 10 = 56. Two jumps of ten land on 56. The 6 ones never change.
Common mistakes: Drawing 2 small hops (38).; Changing the ones digit.
Time for a middle child: about 3 min.

Think harder (page 3): the taught idea in a new setting

T1 **Add 34 + 25. Fill in the blanks.**

(a) Jump: 34 + ___ = ___
(b) Hop: ___ + ___ = ___
(c) So 34 + 25 = ___

Answer: 59 (jump 20 to 54, then hop 5 to 59)

Hint (say this first): First jump 2 tens from 34. Where are you? Then hop 5 ones.
Worked solution: 25 is 2 tens and 5 ones. 34 + 20 = 54. Then 54 + 5 = 59. Check: 3 tens and 2 tens are 5 tens; 4 ones and 5 ones are 9 ones.
Common mistakes: Adding only the tens and stopping at 54.; Adding 2 + 5 as 7 tens (74).
Time for a middle child: about 4 min.

T2 **Subtract 68 - 24. Fill in the blanks.**

(a) Jump back: 68 - ___ = ___

(b) Hop back: ___ - ___ = ___

(c) So 68 - 24 = ___

Answer: 44 (jump back 20 to 48, then hop back 4 to 44)

Hint (say this first): Jump back 2 tens from 68. Then hop back 4.

Worked solution: 24 is 2 tens and 4 ones. 68 - 20 = 48. Then 48 - 4 = 44.

Common mistakes: Taking 24 away from the ones only.; Hopping forward instead of back.

Time for a middle child: about 3 min.

T3 **Both frogs start on 12. Frog A jumps 3 tens, then hops 4. Frog B hops 4, then jumps 3 tens. Fill in a to c.**

(a) Frog A lands on ___.

(b) Frog B lands on ___.

(c) Is the landing number the same for both frogs? Circle Yes or No.

Answer: Yes, both land on 46 (30 and 4 are added in either order)

Hint (say this first): Work out where Frog A lands. Then Frog B. Compare.

Worked solution: Frog A: 12 + 30 = 42, then 42 + 4 = 46. Frog B: 12 + 4 = 16, then 16 + 30 = 46. They land on the same number: adding the same jumps and hops in a different order gives the same total.

Common mistakes: Saying Frog B lands on 16 because the hops came first.; Adding 3 + 4 and landing on 19.

Time for a middle child: about 5 min.

T4 **Ali starts on a number. He jumps forward 3 tens and ends on 78. Where did Ali start?**

Ali started on ___.

Answer: 48 (jump back 3 tens from 78)

Hint (say this first): Go backwards. Undo the jumps: jump back 3 tens from 78.

Worked solution: Undo the 3 jumps of ten: 78, 68, 58, 48. Check: 48 + 30 = 78.

Common mistakes: Adding 30 to 78 (108).; Taking away only 3 (75).

Time for a middle child: about 3 min.

Go further (page 4): mastery and extension

G1 **Mei is on 7 and must get to 97. Each time she can jump 10 or jump 20. Many answers are right. Fill in a and b.**

(a) Write the numbers Mei lands on in one way.

(b) Write the numbers for a different way.

Answer: Any two different routes of 10s and 20s from 7 to 97, e.g. 7, 17, 27, 37, 47, 57, 67, 77, 87, 97 (nine jumps of 10) and 7, 27, 47, 67, 87, 97 (four of 20 and one of 10).

Hint (say this first): The jumps must add up to 90. Check each landing number: it is 10 or 20 more than the one before.

Worked solution: Every jump adds 10 or 20, and all the jumps together add 90 (97 - 7). Examples: nine jumps of 10; or 20, 20, 20, 20, 10 landing on 27, 47, 67, 87, 97. The ones digit stays 7 on every landing.

Common mistakes: Landing on a number that is not 10 or 20 more than the one before.; Two routes that are the same jumps in the same order.; Stopping at 87 or going past 97.

Time for a middle child: about 5 min.

G2 **Now Mei wants the fewest jumps from 7 to 97. She can jump 10 or jump 20. Fill in a and b.**

(a) The fewest number of jumps is ___.

(b) Why can Mei not do it in 4 jumps? I know because ___.

Answer: 5 jumps: four jumps of 20 and one jump of 10 (any order). Four jumps travel at most 80, so she reaches only 87.

Hint (say this first): 90 is 9 tens. A jump of 20 is 2 tens. How many jumps of 20 fit?

Worked solution: She must travel 9 tens. Four jumps of 20 make 8 tens. One jump of 10 makes the last ten. That is 5 jumps. Four jumps can travel at most 80, which only reaches 87, so 5 is the fewest.

Common mistakes: Saying 4 (80 only reaches 87).; Saying 9 (only using jumps of 10).

Time for a middle child: about 6 min.

G3 **Start at 25. Jump forward 3 tens, jump back 1 ten, then jump forward 2 tens. Then hop and land on 69. Fill in a to c.**

(a) Write each number you land on before the hop.
(b) The hop is ___ ones forward.
(c) The three jumps are the same as one jump of ___ tens forward.

> **Answer: 55, 45, 65, then a hop of 4 to 69. The three jumps are the same as 4 tens forward.**

Hint (say this first): Do one move at a time and write where you land each time. Then look at 65 and 69.
Worked solution: 25 + 30 = 55. 55 - 10 = 45. 45 + 20 = 65. To land on 69 the hop is 4. Short way: 3 - 1 + 2 = 4 tens forward, so 25 + 40 = 65.
Common mistakes: Jumping forward on the jump back.; Saying the hop is 44 (69 - 25).; Saying 6 tens for (c) (adding 3 + 1 + 2).
Time for a middle child: about 5 min.

Investigation (page 5): Jump road

Easy first step: Start at 6. Jump in tens 5 times. Write every number you land on.
Then: Start at 6 and keep jumping in tens. Will you land on 50? Write Yes or No. Then try other start numbers from 0 to 50. Which start numbers can land on 50? Write what you notice.

> **Model answer: From 6 you land on 16, 26, 36, 46, 56 and so on: every number ends in 6, so you never land on 50. Jumps of ten never change the ones digit. Only a start number that ends in 0 (0, 10, 20, 30, 40 or 50 itself) can land on 50.**

Look for: A child who tests more than one start number, says the ones digit stays the same, and gives a reason using tens and ones (a jump is whole bars). A child who says start 20 can land on 50 because 20 + 30 = 50.
Common mistakes: Saying you can land on any number.; Counting hops instead of jumps.

Talk and explain (minutes 70 to 76)

Mei says: when you jump forward 2 tens, the ones digit changes. Is Mei right? Use your kit to show why. Sentence starter: I think ... because ...

> **Model answer: No. 34 + 20 = 54, and the 4 ones are still there. Two tens are 2 bars, so only the tens digit changes, from 3 to 5.**

Look for: A reason that uses bars or tens (not just "because I added"). Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **Add.**

46 + 30 = ___

> **Answer: 76**

Worked solution: 4 tens and 3 tens make 7 tens. The 6 ones stay: 76.
Common mistakes: Writing 49 (adding 3 to the ones).; Writing 16.

E2 **Subtract.**

(a) 74 - 30 = ___
(b) 74 - 3 = ___

> **Answer: 44 and 71**

Worked solution: (a) 30 is 3 tens: jump back to 44. (b) 3 is a hop: hop back to 71.
Common mistakes: Doing the same for both parts.; (b) writing 44.

E3 **Mei is on 52. She jumps forward 2 tens, then hops forward 3. Fill in the blanks.**
(a) After the jump she is on ___.
(b) After the hop she is on ___.

Answer: 72 and 75

Worked solution: 2 tens forward from 52 is 72. Then a hop of 3 forward is 75.
Common mistakes: Writing 54 (adding 2 to the ones).; Stopping at 72.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (knowing if it is a jump or a hop; taking away tens; jump, then hop; the number line; something else); *What helped me* (the bars and squares; drawing the jumps; trying again; I asked for help; I was not stuck); *Next time I will try* (drawing it; using objects; checking my answer; asking a question); *I am still stuck (my tutor will come and help)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	A box has 4 bundles of 10 pencils and 5 single pencils. Ali puts in 3 more bundles of 10. How many pencils are in the box now?	75
H2	Start here	Subtract. (a) 86 - 40 = ___ (b) 86 - 4 = ___	46 and 82
H3	Start here	Ravi and Mei each have 53 stickers. Ravi gives away 3. Mei gives away 30. (a) Ravi has ___ stickers left. (b) Mei has ___ stickers left.	Ravi 50, Mei 23
H4	Think harder	Add 36 + 23 in two ways: (a) jump first, then hop; (b) hop first, then jump. Is the answer the same? Circle Yes or No.	59 both ways. Yes, the same.
H5	Go further	(a) Draw a number line from 0 to 100. Put a coin on 7 and jump it in tens. Write each number. (b) What stays the same?	7, 17, 27 ... 97 and 3, 13, 23 ... 93. The ones digit stays the same.

Online practice (`out/l06/practice`, key `t1l06`): 14 items; 11 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to add and take away tens and ones in their head and to say why. This week: jumping in tens and hopping in ones.
Term line (plan, section 5): Your child is learning to see numbers as tens and ones, to notice patterns and to say why. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Tens Jump, Ones Hop** s-title	TODAY WE WILL: / Jump in tens, hop in ones, and say which digit changes.	Lesson 6 of Term 1, 90 minutes. Objective: add and take away tens and ones in your head, jumping the tens first and then hopping the ones. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 28, mini-lesson 28 to 38, brain break 38 to 42, workbook 42 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Nobody is told which page they should be on. Tell the group the lesson is recorded (you and the screen only) and that a child may keep the camera off.
2	2–8	**Warm-up: count in tens** s-rem1	Clap on every number / 10 20 30 40 50 60 70 80 90 100 / Now start at 3 and add ten each time. / 3 13 23 33 43 53 63 73 83 93	Count in tens to 100 with a clap on each number. Then start at 3 and count on in tens. Ask what stays the same (the ones digit, 3). This is a spiral review of skip counting in tens (holiday topic) and lets a new child join: they hear it before they say it.
3	2–8	**Tens and ones** s-rem2	[picture: bundles 4 tens 6 ones] / A bar is a ten. A square is a one. How many tens? How many ones? / 52 70 83	Say each number; children say how many tens and how many ones, with fingers or a whispered answer. Answers: 52 is 5 tens and 2 ones, 70 is 7 tens and 0 ones, 83 is 8 tens and 3 ones. Point to the bars and then the squares in the picture. No writing, no marks.
4	2–8	**Ten and some more** s-rem3	[picture: bundles 4 tens 0 ones] / 40 and 5 more is 45. Count on from the tens. / 30 + 4 50 + 7 80 + 2	Answers: 34, 57, 82. Say the tens word slowly, then count on the ones: forty... forty-one, forty-two. This is the bridge to jumping and hopping. Show the bars as a number path if a child is unsure.
5	8–12	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / Mei has these sticks. / Then she gets another bundle of 10 sticks. / How many sticks does Mei have now? / [picture: bundles 2 tens 6 ones] / PUZZLE 2: think: / Ravi says 43 + 20 = 45. / Is Ravi right? Write yes or no, and why.	Puzzle 1 (26 sticks and one more bundle) is an access question: any child can try it by counting or drawing. Puzzle 2 is a thinking question on the lesson's key idea: 20 is 2 tens. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Everyone holds the paper to the camera once, at the same moment, after puzzle 2; there is no private chat and no board in front of the group. Answers: 36; no, 63. Do not reveal Puzzle 2 until Talk time. Do not write scores or keep a list of who found it easy.
6	12–28	**Explore: jump with bars** s-explore1	Build 34: 3 bars and 4 squares. / A big jump is a bar. Add one bar. Say the number. / Jump again. Say the number. / Jump back: take a bar away. Say the number. / [picture: bundles 3 tens 4 ones]	Round 1, about 5 minutes. Objects first, no writing. A bar is a big jump of ten. Children build 34, jump forward twice (44, 54), then back (44, 34, then 24). Hold the build to the camera on a count of three. Ask which digit changed and which stayed the same. A child with too few bars can use the full kit page: 9 bars and 20 squares are enough for this lesson.
7	12–28	**Explore: hop with squares** s-explore2	Build 24: 2 bars and 4 squares. / A small hop is a square. Hop forward 3 times. / Hop back 2. / Which digit changed? / [picture: bundles 2 tens 4 ones]	Round 2, about 3 minutes. Hop three squares forward: 27. Hop two back: 25. The ones digit changes and the tens digit stays. Say it in the children's words: a jump changes the tens, a hop changes the ones.
8	12–28	**Explore: jump and hop** s-explore3	Build the first number. Jump the tens. Then hop the ones. Hold up your kit when I count to three. / 34 + 25 41 + 38 68 - 24	Round 3, about 5 minutes. 34 + 25: 2 bars then 5 squares gives 59. 41 + 38: 3 bars then 8 squares gives 79. 68 - 24: take away 2 bars then 4 squares gives 44. Check each build as the children hold it to the camera. Say each answer aloud for the recording. A child who adds the 2 as ones is mixing up jump and hop: ask how many squares one bar is worth.

9	12–28	**Draw the jumps and hops** s-explore4	Draw a dot. Write the start number. / Draw a big arc for the jump. Write +20. / Write where you land. / Draw a small arc for the hop. Write +5. Write where you land. / [picture: number_line]	Round 4, about 5 minutes. Draw on paper as the children draw: dot, arc, landing number. A big arc is a jump of tens, a small arc is a hop of ones. Draw 34 to 59 first, then 68 back to 44 with arcs going back. The paper line is the picture children use in question 7 and in Think harder.
10	12–28	**What did we notice?** s-explore5	Which one changes the tens digit, a jump or a hop? / Which one changes the ones digit? / Does it matter if you hop first or jump first? / How many ones can we hop before we reach ten?	Take one or two answers from volunteers. Key ideas: a jump of ten changes the tens digit; a hop of one changes the ones digit; jumping first or hopping first lands in the same place (do not push the proof, the Think harder page tests it); we stay inside a ten today. Repeat a child's answer in your own words so the recording picks it up.
11	28–38	**Jump forward: add tens** s-m1	[picture: number_line] / Each jump adds 1 ten. The ones digit stays the same. / 34 + 20 = 54	Draw the number line on paper as you talk. The arcs are the jumps. Two jumps of ten add 20. Ask: which digit changed? Answer: the tens digit, 3 to 5. The 4 ones stay.
12	28–38	**Jump back: take away tens** s-m2	[picture: number_line] / Each jump back takes away 1 ten. The ones digit stays the same. / 54 - 20 = 34	The same picture backwards. Arcs go back, so the numbers get smaller. Say: 54 take away 10 is 44, take away 10 again is 34. Link to the kit: take two bars away.
13	28–38	**Hop: ones** s-m3	[picture: number_line] / 34 + 3 = 37 / [picture: number_line] / 37 - 2 = 35 / A hop changes the ones digit. The tens digit stays.	Small hops of one. Show a hop as a small arc and a jump as a big arc. Ask: is five a jump or a hop? Answer: a hop, it is ones. Keep every example inside a ten: no hop may pass nine ones today.
14	28–38	**Jump, then hop: add** s-m4	[picture: number_line] / 25 is 2 tens and 5 ones. Jump the tens first. Then hop the ones. / 34 + 25 = 59	Worked example. Split 25 into 20 and 5. Jump 20 from 34 to 54, then hop 5 to 59. Ask: how many tens and how many ones in 36? (3 tens, 6 ones) to check children can split a number. The example stays inside a ten (4 + 5 = 9). Do not mention carrying or renaming.
15	28–38	**Jump, then hop: take away** s-m4b	[picture: number_line] / 25 is 2 tens and 5 ones. Jump back the tens first. Then hop back the ones. / 59 - 25 = 34	Worked example. 59 - 25: jump back 20 to 39, then hop back 5 to 34. The arcs go backwards, so the numbers get smaller. Ask: what is 87 - 32? Answer: jump back 3 tens to 57, hop back 2 to 55. Stay inside a ten (9 - 5 = 4, 7 - 2 = 5).
16	28–38	**Your turn: write the answer** s-m5	Write on your paper. Timer: 1 minute / 45 + 30 78 - 20 64 - 3 52 + 36	Each child writes on their own paper. Walk the cameras. Do not ask children to hold boards up in front of the group. Children may draw the jump or use the kit. Answers are on the next slide.
17	28–38	**Check: write the answer** s-m5a	45 + 30 = 75 78 - 20 = 58 64 - 3 = 61 52 + 36 = 88	Answers: 75, 58, 61, 88. Read each answer aloud so a child watching the recording can mark their own work. Ask which were jumps, which were hops and which needed both (the third is a hop, the fourth is a jump then a hop).
18	28–38	**Jump or hop?** s-m6	30 6 50 2 / Tens are a jump. Ones are a hop.	Quick-fire. Say a number and ask: jump or hop? Answers: 30 jump, 6 hop, 50 jump, 2 hop. A multiple of ten is a jump because it is whole bars. Keep it quick: this is the decision children make on the workbook pages.
19	38–42	**Jump the tens, hop the ones** s-break	A big jump for every ten. A small hop for every one. / START ON 34: / Jump forward 2 times. / Hop forward 3 times. / Jump back 1 time. / Hop back 4 times. / Say the number each time. Sip some water.	Four minutes. Stand up on camera with a little clear space; no stairs and no running. Start on 34: jump forward twice (44, 54), hop forward three times (55, 56, 57), jump back once (47), hop back four times (46, 45, 44, 43). Count the numbers aloud together. Keep it gentle for small spaces. No equipment.
20	42–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. Timer: 9 minutes / questions S1, S2, S3	Read questions 1 to 3 aloud, including the sums in question 3. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder.

21	42–70	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children mark their own page in a different colour. Never ask a child to show a wrong answer to the group.
22	42–70	**Workbook page 2: Start here** s-ws2	Questions 4 to 7. Timer: 9 minutes / questions S4, S5, S6, S7	Read all four questions aloud with their numbers. Question 6 is a spot-the-mistake: Ben hopped when she should have jumped. Question 7 asks children to draw jumps on a number line in the box: show them on paper how to draw a dot, an arc and +10. Children may use the kit.
23	42–70	**Check page 2** s-ws2ans	answers S4, S5, S6, S7	Read each answer aloud so a child watching the recording can mark their own work. Question 6 answer: no, 65, because 20 is 2 tens, not 2 ones. Never ask a child to show a wrong answer to the group.
24	42–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Jump and hop in a new setting / Go further (page 4) | Bigger journeys on the number line / Investigation (page 5) | Always, sometimes or never? / Check | Finish and check your own work / Everyone may choose. You can swap at any time.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 4 minutes.
25	70–76	**Talk and explain** s-talk	MEI SAYS: / When you jump forward 2 tens, the ones digit changes. / Is Mei right? Use your kit to show why. / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Then volunteers first; a pass is always fine; never call a child by name to answer. Model answer: no. 34 + 20 = 54, and the 4 ones are still there. Two tens are two bars, so only the tens digit changes. Repeat each child's answer in your own words for the recording.
26	70–76	**Back to the sneak-peek** s-sneakans	[picture: number_line] / No. 43 + 20 = 63. Twenty is 2 tens, so we jump.	Reveal puzzle 2 now. Say: many of you wrote a good reason. Ravi hopped 2 ones (45) instead of jumping 2 tens (63). Do not read out anyone's earlier answer.
27	76–82	**Exit ticket** s-exit	On your own. No looking back. Timer: 3 minutes / questions E1, E2, E3	Children answer on page 5. After the timer everyone holds the ticket to the camera at the same moment, so only the tutor sees it (no private chat). Stamp the inquiry journal.
28	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children check their own ticket in a different colour.
29	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, check in during the open session next lesson (a quick visit to the shared room or a hand signal, parent nearby); never a direct message. The Next time I will try row is optional.
30	85–90	**Well done, inquirers!** s-end	Show a grown-up one thing you found out today. / NEXT LESSON: / Take Away, Count Up. Bring your kit and a pencil.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (stamps, the look-back). Do not start anything new. If the whole class is ahead, show the optional menu slide and use the Spare-time menu from the tutor guide.
31	optional	**Go further: what if?** s-opt	Start at 5. Jump in tens. What is the last number you land on before 100? / How many jumps is that? / [picture: number_line]	Optional, only if you are on time. Answer: 95, after 9 jumps (5, 15, 25 … 95). One more jump would pass one hundred, which we meet in the next lessons. Take two answers and move on.

32	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Add a ten to 34, 67 and 90. What changes? / Make a puzzle | Write a jump and hop story for your grown-up / Teach it back | Show how to work out 52 + 31 with a jump and a hop / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your kit of bars and squares. Pause or rewind any time.*	Gather your workbook, pencil, paper and kit. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Say the numbers out loud. Pause if you need longer.*	Play along. Say each number before the video does.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause each time I ask. Build the number with your bars and squares, then press play to check.*	Pause. Build each number with your kit. Write what you notice.
28:00	Mini-lesson	*Recording: keep watching and copy the jumps and hops onto your paper. Rewind if you miss a step.*	Watch and copy the jumps and hops. Rewind any step you missed.
38:00	Brain break	*Recording: stand up and join in! Big jumps for tens, small hops for ones.*	Stand up and jump the tens and hop the ones.
42:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up your answer to the talk question, and say why.*	Pause. Explain your answer to someone at home, with a reason.
76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one thing you found out.*	Show a grown-up your workbook and tell them one thing you found out.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:34	1:47	ok
remember	6:00	5:24	6:24	ok
sneak	4:00	3:25	4:00	ok
explore	16:00	12:50	15:12	ok

mini	10:00	8:27	9:59	ok
break	4:00	3:38	4:20	ok
work	28:00	26:28	31:34	ok
talk	6:00	5:15	6:15	ok
exit	6:00	4:32	5:23	ok
look	3:00	2:28	2:55	ok
flex	5:00	0:38	0:44	ok
Total	90:00	74:38	88:33	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 7 questions; a middle child needs about 8.0 min for page 1 and 11.0 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.